Sardor Sobirov
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https://sardors-portfolio.vercel.app

Education

University of California San Diego (UCSD) La Jolla, CA
Bachelor of Science in Data Science *Jun 2026*

- **Relevant Coursework**: Machine Learning Algorithms, Deep Learning and Neural Networks, Probabilistic AI and Graphical Models, Data Mining and Statistical Learning, Distributed Data Systems and Scalable Analytics.

Skills

Languages: Python (pandas, NumPy, scikit-learn, PyTorch), SQL, JavaScript, HTML/CSS.
Data & Analytics: A/B Testing, Hypothesis Testing, Funnel Analysis, KPI Design, Metric Trees, Exploratory Data Analysis.
Data Platforms & Cloud: BigQuery, Databricks, AWS, PostgreSQL.
Visualization: Tableau, Power BI, Matplotlib, Plotly, ArcGIS.

Experience

Cotality Sep 2025 – Mar 2026
HDSI Research and Development Intern *San Diego, CA*

- Developed a React/TypeScript geospatial dashboard using MapLibre GL, D3.js, Vite, and Tailwind CSS for Cotality's data science team to investigate county-level property-characteristic distributions, wildfire risk signals, across California.
- Used BigQuery GIS, SQL, H3 spatial indexing, spatial joins, buffers, and centroid-based geometry processing to prepare property and land-cover data for county-level anomaly detection.

Apptics.ai Jun 2025 – Sep 2025
Data Engineer Intern *Walnut Creek, CA*

- Built a reproducible Python ETL pipeline ingesting daily paid media performance data from the Meta Ads API into PostgreSQL for 5 DTC brands, consolidating 1K-10K/day per brand spend into a single source of truth for reporting.
- Designed an analytics-ready relational schema (campaign/ad set/ad levels) and standardized 25+ marketing KPIs, using SQL and PowerBI, enabling consistent cross-brand performance tracking toward higher-converting ad sets.
- Ran A/B tests on Meta ad creatives using Python and Tableau, then translated statistical findings into plain-language recommendations for non-technical marketing teams contributing to a 35% improvement in acquisition efficiency.

United States Department of Defense Sep 2024 – Dec 2024
Research and Development Intern *La Jolla, CA*

- Conducted 50+ stakeholder interviews using Lean LaunchPad methodology to translate ambiguous user needs into a prioritized product roadmap, ensuring R&D investments were tied to concrete operational use cases.
- Designed a portfolio analytics dashboard in Excel and Power BI mapping technology maturity signals to DoD use cases; presented findings to cross-functional leadership to inform multi-million dollar transition decisions.

Projects

Spatiotemporal Modeling Agricultural Resilience Interactive Dashboard

- Built an end-to-end geospatial ML pipeline for the 13-state Corn Belt, integrating USDA CDL, MODIS NDVI, and NASA SMAP data via rasterio, xarray, GeoPandas, JAX, NetCDF, Parquet, and YAML-driven workflows.
- Modeled crop phenology, rotation, soil-moisture anomalies, and crop prediction using HSGP, Dirichlet-Multinomial, NIG Bayesian scoring, and LightGBM; achieved 79.2% crop-classification accuracy with SHAP and ablation diagnostics.

Bayesian Spatial Transit Accessibility Interactive Dashboard

- Developed a Bayesian spatial transit-equity model for 720 San Diego census tracts using GTFS, ACS, OSM/r5py, GeoPandas, PyMC BYM2, ICAR effects, and posterior risk estimation.
- Built a Next.js/TypeScript, Deck.gl, and MapLibre dashboard comparing Bayesian vs. deterministic intervention targeting; identified 178 probable transit deserts and improved top-20 targeting efficiency by 1.5x.

Probabilistic Hurricane Track Forecasting Interactive Dashboard

- Built a probabilistic hurricane track forecasting pipeline on NOAA IBTrACS data, processing 721K+ observations across 13K+ storms with Python, NumPy, pandas, SciPy, and Kalman filtering.
- Engineered storm-relative geospatial features and evaluated open-loop forecasts across 6–72 hour horizons, benchmarking adaptive Kalman filters against a persistence baseline to diagnose model limits and forecast uncertainty.

Bayesian Detection for EEG Regime Shifts Interactive Dashboard

- Implemented Bayesian Online Change-Point Detection for THINGS-EEG using Python, SciPy, Welch PSD, log-bandpower features, and numerically stable log-space inference across 10 participants.
- Validated EEG regime-shift detection with synthetic tests and neural features, achieving 91% detection on large mean shifts and showing variance-based features captured within-epoch changes missed by mean amplitude.